

# 1er Parcial 2021

domingo, 7 de noviembre de 2021

21:10

$$a) G_R(s) = 0.5 \left( 1 + \frac{1}{s} \right) = 0.5 \frac{(s+1)}{s}$$

$$t_e \leq 4 \text{ seg} \rightarrow T \leq \frac{t_e}{10} = 0.4 \text{ seg.}$$

$$G_R(z) = Z[G_R(s)] = Z\left[0.5 \left( 1 + \frac{1}{s} \right)\right] = 0.5 + \frac{0.5z}{z-1}$$

$$\boxed{G_R(z) = \frac{z-0.5}{z-1}} \quad 1 \text{ pto}$$

$$b) G_R(z) = \left( \frac{z-1}{z} \right) \cdot Z\left[0.5 \left( \frac{1}{s} + \frac{1}{s^2} \right)\right] = 0.5 \frac{(z-1)}{z} \left( \frac{z}{z-1} + \frac{Tz}{(z-1)^2} \right) =$$

$$= 0.5 + \frac{0.5 \cdot 0.4}{(z-1)} = \frac{0.5z - 0.5 + 0.2}{(z-1)} = \frac{0.5z - 0.3}{z-1} = \frac{0.5(z-0.6)}{(z-1)} \quad 1 \text{ pto}$$

$$c) G_R(z) = \frac{1 - 0.5z^{-1}}{1 - z^{-1}} = \frac{U(z)}{E(z)} \rightarrow \boxed{U_k = U_{k-1} + e_k - 0.5e_{k-1}}$$

$$G_R(z) = \frac{0.5 - 0.3z^{-1}}{1 - z^{-1}} = \frac{U(z)}{E(z)} \rightarrow \boxed{U_k = U_{k-1} + 0.5e_k - 0.3e_{k-1}} \quad 1 \text{ pto}$$

$$d) \{e_k\} = \{0, 1, 0, 0, \dots\} \rightarrow E(z) = A \cdot z^{-1}$$

$$U(z) = G_R(z) \cdot E(z) = \frac{z-0.5}{z-1} \cdot A \cdot z^{-1} = A \frac{z \cdot z^{-1}}{z-1} - 0.5 \cdot A \frac{z \cdot z^{-2}}{z-1}$$

$$\{U_k\} = A \cdot \{0, 1, 1, 1, \dots\} - 0.5 \cdot A \cdot \{0, 0, 1, 1, 1, \dots\} =$$

$$\{U_k\} = \{0, A, 0.5A, 0.5A, 0.5A, \dots\}$$

$$\text{Valor máximo } A \leq 10 \rightarrow \boxed{A_{\max} = 10.} \quad 1 \text{ pto}$$

$$U(z) = \frac{0.5z - 0.3}{z-1} \cdot A \cdot z^{-1} = 0.5 \cdot A \frac{z \cdot z^{-1}}{z-1} - 0.3 \cdot A \frac{z \cdot z^{-2}}{z-1}$$

$$\{U_k\} = 0.5 \cdot A \{0, 1, 1, 1, \dots\} - 0.3 \cdot A \{0, 0, 1, 1, 1, \dots\} =$$

$$\{U_k\} = \{0; 0.5A; 0.2A; 0.2A; 0.2A, \dots\}$$

$$(U_k)_{\max} = 0.5A \leq 10 \rightarrow A \leq 20 \rightarrow \boxed{A_{\max} = 20.} \quad 1 \text{ pto}$$